MATH 372: COMBINATORICS
HOMEWORK 05

DUE MONDAY, 16 NOVEMBER, 10P.M., VIA GRADESCOPE (MOVED FROM NOVEMBER 13)

Remark. **When you can start.** Problems 1–3 use the labelled constructions (November 2 and 4) and Pólya's theorem (November 6 and 9). Problem 4 uses transfer matrices and partial fractions (November 11 and 13); it is short, and it is the only problem on that material this term, since there is no problem set 6.

Remark. **Conventions.** A *labelled* class has exponential generating function $A(z) = \sum_\alpha z^{|\alpha|}/|\alpha|!$, so that $a_n = n!\,[z^n]A(z)$. For a labelled class $\mathcal{B}$ with no object of size 0, the constructions and their translations are

$$\mathrm{SEQ}(\mathcal{B}) \mapsto \frac{1}{1-B(z)}, \qquad \mathrm{SET}(\mathcal{B}) \mapsto e^{B(z)}, \qquad \mathrm{CYC}(\mathcal{B}) \mapsto \log\frac{1}{1-B(z)},$$

and $\mathrm{SET}_{\geq 1}$, $\mathrm{CYC}_{\mathrm{odd}}$, and so on, restrict the number or the sizes of the components in the obvious way. The *cycle index* of a group G acting on $[n]$ is $Z_G(p_1, p_2, \dots) = \frac{1}{|G|}\sum_{g\in G}\prod_i p_i^{c_i(g)}$, where $c_i(g)$ is the number of i-cycles of g, and Pólya's theorem says that colorings of $[n]$ up to G, counted by the number of each color, have generating function $Z_G(p_i \mapsto x_1^i + \cdots + x_r^i)$. You may use Newton's binomial theorem, $(1+x)^\alpha = \sum_k \binom{\alpha}{k} x^k$.

Remark. **On write-ups.** Each problem is worth up to five points for mathematical content and up to two for writing. Write for another student in this course: say what you are going to show before you show it, justify each step, and use complete sentences. Whenever you give a specification, say in words what the objects are before writing the generating function. End your submission with a short *Methods and tools* note: what software you used and for what, and how you checked anything a machine produced for you.

Problem 1 (parity). (a) Let $\mathcal{V}$ be the class of set partitions of $[n]$ all of whose blocks have *even* size. Write a specification for $\mathcal{V}$, find its exponential generating function in closed form, and compute v_n for $n \leq 6$. Check v_4 by listing.

(b) Let a_n be the number of permutations of $[n]$ all of whose cycles have even length, and b_n the number all of whose cycles have odd length. Find both exponential generating functions in closed form, and show that $a_n = b_n$ whenever n is even. (You will find that the two generating functions differ by a factor of $1 + z$; say why that proves the claim.)

(c) Compute a_n and b_n for $n \leq 4$ by listing cycle types, and check (b).

Problem 2 (Stirling numbers of the second kind). Let $S(n,k)$ be the number of set partitions of $[n]$ into exactly k blocks, so that, as in class, $\sum_n S(n,k)\frac{z^n}{n!} = \frac{(e^z-1)^k}{k!}$.

(a) Expand $(e^z-1)^k$ by the binomial theorem and extract the coefficient of $z^n/n!$ to prove

$$S(n,k) = \frac{1}{k!}\sum_{j=0}^{k}(-1)^j\binom{k}{j}(k-j)^n.$$

(b) Use (a) to compute $S(5,2)$ and $S(5,3)$, and check them against a direct count.

(c) Let $B(z) = e^{e^z - 1}$ be the exponential generating function of the Bell numbers B_n (all set partitions of $[n]$). Differentiate to show $B'(z) = e^z B(z)$, and deduce the recurrence

$$B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_k.$$

Then explain the recurrence directly, without generating functions, by looking at the block containing the element $n+1$.

Problem 3 (necklaces and bracelets). A *necklace* is a coloring of the n vertices of a regular n-gon up to rotation; a *bracelet* is a coloring up to rotation and reflection. Take $n = 8$ and two colors, black and white.

(a) Write the cycle index of the rotation group C_8 acting on the 8 vertices. (Sort the eight rotations by their order.) How many necklaces are there?
(b) Write the cycle index of the dihedral group D_8 of order 16. (There are two kinds of reflection; say what each does to the vertices.) How many bracelets are there?
(c) Count necklaces by the number of black beads: substitute $p_i \mapsto 1 + q^i$ into (a) and expand. Check two coefficients by listing the necklaces with that many black beads.
(d) The polynomial in (c) is palindromic. Say why, in one sentence, for every n.

Problem 4 (a transfer matrix, and a callback). Let w_n be the number of words of length n over the alphabet $\{a, b, c\}$ in which no two consecutive letters are equal.

(a) Write the transfer matrix A (rows and columns indexed by the three letters; the (x, y) entry is 1 if y may follow x) and express w_n in terms of powers of A. Find w_n in closed form.
(b) Now count *circular* words: colorings of the vertices of the n-cycle C_n with three colors so that adjacent vertices differ, with the vertices labelled (so no symmetry is divided out). Explain why the number is $\operatorname{tr} A^n$, compute the eigenvalues of A, and conclude that the count is $2^n + 2(-1)^n$.
(c) Write the generating function $\sum_{n \geq 1} (\operatorname{tr} A^n) z^n$ as a rational function, and read off its partial-fraction decomposition. Which pole controls the growth rate, and what is the growth rate?
(d) You have met the matrix A before: say what graph it is the adjacency matrix of, and what Act I theorem (b) is a special case of.

Bonus Problem (unordered binary trees). Let $\mathcal{T}$ be the class of rooted labelled trees in which every node has at most two children, with *no* order on the children: $\mathcal{T} = \mathcal{Z} \star \mathrm{SET}_{\leq 2}(\mathcal{T})$, so $T(z) = z\left(1 + T(z) + \frac{1}{2}T(z)^2\right)$. Use Lagrange inversion, $[z^n]T = \frac{1}{n}[u^{n-1}]\phi(u)^n$ with $\phi(u) = 1 + u + \frac{1}{2}u^2$, to compute $n!\,[z^n]T$ for $n \leq 5$, and check $n = 3$ by drawing the trees.